Predicting RNA-Protein Interactions From Sequence Alone

RNAInterAct and RPIembeddor: a homology-aware dataset and a compact transformer that generalizes to unseen RNA families (GEM workshop, ICLR 2024)

Non-coding RNAs help run the cell largely by binding proteins, yet mapping those interactions in the lab with assays like SELEX and CLIP-seq is slow and costly. Computational predictors exist, but most sidestep the general question by training one model per protein, which needs a large interaction dataset for that specific protein. Such datasets exist for only a few hundred of the roughly 2,000 human RNA-binding proteins.

What the field lacks is a predictor that answers a simpler-sounding but harder question: given any RNA and any protein, and nothing but their sequences, do they interact? Two recent ideas make it plausible. Learning across many interaction types at once, rather than one protein at a time, helps when labels are scarce; and foundation models trained on huge unlabeled biological corpora expose structural and functional signal that raw sequences hide.

This work, from the University of Freiburg and the ELLIS Institute Tuebingen, delivers both halves of the missing piece: RNAInterAct, a large curated dataset of non-coding RNA-protein interactions with homology-separated splits, and RPIembeddor, a compact transformer of just 1.4M parameters that classifies interactions from sequence embeddings alone and, unlike prior tools, generalizes to RNA families entirely absent from its training set.

Paper (bioRxiv preprint): https://www.biorxiv.org/content/10.1101/2024.11.08.622607v1 . Dataset (the RNAInterAct archive): https://ml.informatik.uni-freiburg.de/research-artifacts/RNAInterAct/RNAInterAct.tar.gz . Both are released by the authors for open research use.

Why sequence-only, family-general prediction is hard

Prior methods such as IPMiner and XRPI can score well when training and test sets share RNA families, because sequence homology leaks information between the two. The real test is generalization: can a model trained on some RNA families classify interactions for families it has never encountered? RPIembeddor is built and evaluated for exactly that setting, and its dataset is split so that no RNA family appears in both training and testing.

RPIembeddor: two foundation models, one classifier

The model leans on two pre-trained foundation models. RNA sequences are embedded with RNA-FM; protein sequences with the 150M-parameter ESM-2, chosen over structure predictors like AlphaFold because it needs no multiple sequence alignments. Both produce 640-dimensional embeddings that carry structural and functional priors the raw letters do not expose.

As Figure 1 shows, two parallel feed-forward layers normalize the two embeddings to a common size, and attention-based encoder layers then process the RNA and protein representations symmetrically, so neither modality dominates. The latent representations are concatenated, passed through further feed-forward layers, and a final linear layer with a sigmoid outputs the probability that the pair interacts. The whole classifier holds only 1.4M parameters and trains with a binary loss and the AdamW optimizer.

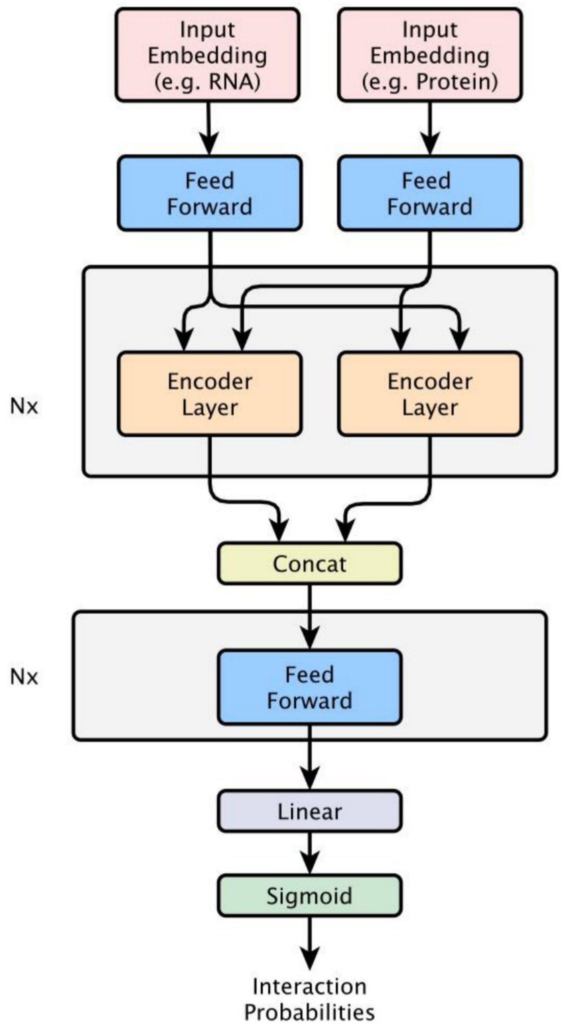

Figure 1. RPIembeddor architecture. RNA and protein embeddings are normalized by parallel feed-forward layers, processed symmetrically through attention-based encoder layers so each modality has equal influence, then concatenated and passed through a final linear layer with a sigmoid to output an interaction probability.

RNAInterAct: a homology-aware benchmark

A model is only as trustworthy as the data it is tested on. The authors build RNAInterAct from the RNAInter database, recovering sequences by cross-referencing NCBI, UniProt, and Ensembl, and annotating RNA families with Rfam and protein clans with Pfam. Those family and clan annotations are what let them generate biologically plausible negatives instead of random pairs.

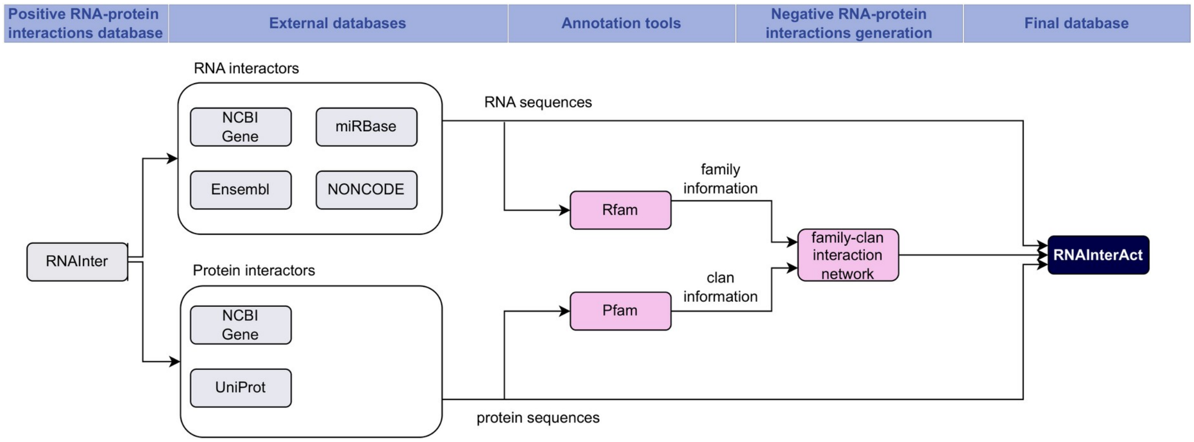

Figure 2. The RNAInterAct curation pipeline: positive interactions from RNAInter are matched to sequences in external databases (NCBI, UniProt, Ensembl, miRBase, NONCODE), annotated with Rfam families and Pfam clans, and combined through a family-clan interaction network to generate biologically meaningful negatives.

The finished dataset holds 122,217 non-coding RNA-protein interactions at a 1:2 positive-to-negative ratio. Crucially, it is split by RNA family into a training set, TRinter (109,214 interactions over 976 families), and a test set, TSfam (13,003 interactions over 172 families), with no family shared between them. Models are additionally tested on the external, positives-only RPI2825 set (871 interactions), which probes generalization across data distributions rather than within one.

Results: real signal where baselines sit at chance

On the homology-separated TSfam test set, the gap between learning and memorizing becomes stark. A model that merely exploited homology would collapse here, because no test family was ever seen during training.

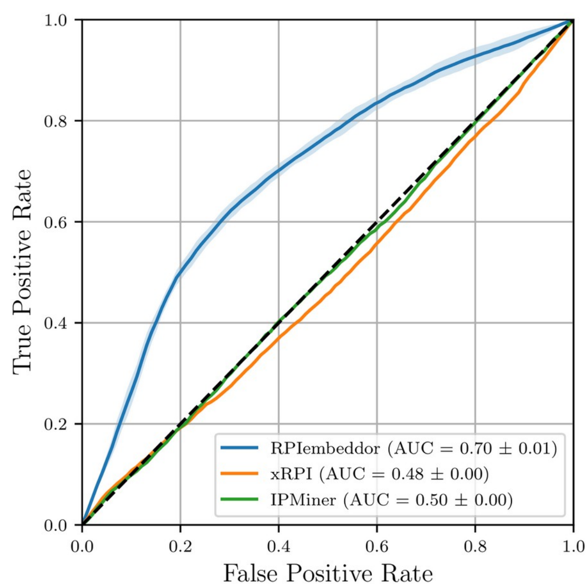

Figure 3. ROC curves on TSfam. RPIembeddor reaches an AUC of 0.70, while XRPI (0.48) and IPMiner (0.50) track the diagonal, meaning they perform at roughly chance on unseen RNA families.

RPIembeddor reaches an F1 of 0.586 and an accuracy of 0.667, with a ROC AUC of 0.70, while XRPI and IPMiner land at AUC 0.48 and 0.50, at or below chance. In concrete terms, RPIembeddor correctly labels 2,971 of 4,887 positive and 5,586 of 8,116 negative TSfam interactions, whereas XRPI collapses to predicting roughly 91% of pairs as interacting even though about 62% are negatives.

Table 2. Mean performance on the homology-separated TSfam test set. RPIembeddor leads on precision-balanced F1 and accuracy, while the baselines trade precision for recall or the reverse.

Model	Prec.	Rec.	F1	Acc.
RPIembeddor	0.550	0.627	0.586	0.667
IPMiner	0.357	0.375	0.366	0.512
XRPI	0.375	0.909	0.531	0.398

On the external RPI2825 set, RPIembeddor also posts the second-best F1 (0.8), evidence that the model transfers across data distributions and is not tuned to a single benchmark.

Ablation: both embeddings are load-bearing

Do the foundation-model embeddings actually carry the signal, or is the attention network doing the work on its own? The authors retrain RPIembeddor after replacing the RNA-FM embedding, the ESM-2 embedding, or both, with random vectors or one-hot encodings, then measure what survives.

Table 3. Ablation on TSfam. Removing either embedding collapses the model into a negative-only classifier; only the full two-embedding model classifies correctly.

Model	Prec.	Rec.	F1	Acc.
RPIembeddor	0.563	0.659	0.605	0.678
One-Hot	0.0	0.0	0.0	0.624
Random-Protein	0.0	0.0	0.0	0.624
RNA-Random	0.0	0.0	0.0	0.624

Every variant collapses into a negative-only classifier: precision, recall, and F1 fall to 0.0, and accuracy sticks at 0.624, exactly the fraction of negatives in the test set. Only with both RNA-FM and ESM-2 embeddings present does the model reach F1 0.605 and accuracy 0.678. The two foundation-model embeddings are not optional add-ons; they are where the structural and functional information lives.

Takeaway

The lasting message is that general, sequence-only RNA-protein interaction prediction becomes achievable when you stand on the shoulders of foundation models. A compact attention network fed with RNA-FM and ESM-2 embeddings outperforms specialized tools and, unlike them, generalizes to RNA families it has never seen. Just as important, the homology-separated RNAInterAct dataset gives the community a benchmark that measures generalization rather than homology leakage. The authors flag clear next steps: adding RNA-structure models and lifting the current sequence-length limit.